1️⃣ Identify the limits of the model used to solve a problem (Item 3)

Relevance: ★★★★★ (Most critical)

Description:

> Identify the limits of the model used to solve a problem. interpret

The key idea of the task is recognizing that:

An average does not determine the exact value for each individual year.

For example:

- 315 days represents the **average across 10 years**.
- Individual years may be **higher or lower than 315**.

Students must recognize the **limitations of the average-value model**, including:

- It cannot determine the exact value for a specific year.
- It only describes the overall trend.

Thus, this competency is the **most central ability required in this task**.

2️⃣ Identify whether a mathematical result makes sense in context (Item 2)

Relevance: ★★★★★

Description:

> Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem. interpret

Students must judge whether the three statements are logically valid.

For example:

Statement 1

"At least one year had exactly 315 days."

→ This **cannot be guaranteed**, so it **cannot be deduced**.

Statement 2

"One year could have been used for more than 315 days."

→ This is **possible**.

Statement 3

"One year might have had zero usage."

→ This is also **theoretically possible**.

Students must evaluate whether these conclusions **make sense given the meaning of an average**.

3️⃣ Recognise the extent and limits of mathematical concepts (Item 8)

Relevance: ★★★★☆

Description:

> Recognise the extent and limits of mathematical concepts and mathematical solutions.
> interpret

Students must understand the **scope of the concept of an average**.

An average can represent:

- An **overall trend**

But it cannot determine:

- The **exact value of each individual observation**

Therefore, students must recognize the **limits of the mathematical concept**.

4️⃣ Interpret a mathematical result back into the real-world context (Item 1)

Relevance: ★★★★☆

Description:

> Interpret a mathematical result back into the real-world context. interpret

Students must interpret the meaning of:

315 days per year

in the real-world context.

This means:

- It represents an **average yearly usage**,
- It **does not imply that every year had exactly 315 days of use**.

Thus, students must translate **a mathematical result into contextual meaning**.

5️⃣ Understand the relationship between context and mathematical representation (Item 9)

Relevance: ★★★☆☆

Description:

> Understand the relationship between the context of the problem and representation of the mathematical solution. interpret

Students must understand that:

- "315 days per year" is a **statistical representation**.
- It does **not necessarily reflect the actual value in each year**.

Understanding the relationship between **real-world context and mathematical representation** is essential to solving the task.

6️⃣ Construct and communicate explanations and arguments (Item 7)

Relevance: ★★★☆☆

Description:

> Construct and communicate explanations and arguments in the context of the problem. interpret

Students must not only answer **Yes or No**, but also understand the reasoning behind their decisions, such as:

- An average cannot determine the value of a single year.
- Extreme values are theoretically possible.

This involves **logical reasoning and explanation**.

1️⃣ Identify the limits of the model used to solve a problem（第3项）

相关性：★★★★★（最核心）

能力描述：

> Identify the limits of the model used to solve a problem. interpret

本题的关键是理解：

平均值并不能确定每一年的具体情况。

例如：

- 315 天只是 **10年的平均值**
- 每一年可能高于或低于 315

学生需要识别 **平均值模型的局限性**：

- 无法确定某一年具体数值
- 只能说明总体情况

因此该能力是本题最核心能力。

2️⃣ Identify whether a mathematical result makes sense in context（第2项）

相关性：★★★★★

能力描述：

> Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem. interpret

学生需要判断三个陈述是否合理：

例如：

陈述1

“某一年正好 315 天”

→ 不能确定，因此 **不能推出**

陈述2

“某一年可能超过315天”

→ 可能成立

陈述3

“某一年可能为0天”

→ 理论上也可能

学生必须判断这些结论是否符合平均值的现实意义。

3️⃣ Recognise the extent and limits of mathematical concepts（第8项）

相关性：★★★★☆

能力描述：

> Recognise the extent and limits of mathematical concepts and mathematical solutions.
> interpret

学生需要理解 **平均值概念的适用范围**：

平均值只能表示：

- 总体趋势

而不能表示：

- 每个个体情况

因此需要理解 **数学概念的适用范围和限制。**

4️⃣ Interpret a mathematical result back into the real world context（第1项）

相关性：★★★★☆

能力描述：

> Interpret a mathematical result back into the real world context. interpret

学生需要将：

315 days per year

重新解释为现实含义：

- 每年使用天数的平均值
- 并不代表每一年都是315

这是 **数学结果 → 现实解释** 的过程。

5️⃣ Understand the relationship between context and mathematical representation（第9项）

相关性：★★★☆☆

能力描述：

> Understand the relationship between the context of the problem and representation of the mathematical solution. interpret

学生需要理解：

- “315 天/年” 是 **统计平均表示**
- 并不表示真实每年的情况

理解 **现实情境与数学表示之间的关系** 是解决问题的重要步骤。

6️⃣ Construct and communicate explanations and arguments（第7项）

相关性：★★★☆☆

能力描述：

> Construct and communicate explanations and arguments in the context of the problem. interpret

学生不仅需要判断 **Yes / No**，还需要理解判断的理由，例如：

- 平均值不能确定单个年份
- 可能存在极端情况

这涉及 **逻辑解释能力。**